

SE LAB-1

Requirements Engineering & UML Use-Case Modelling

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PROBLEM STATEMENT #16 | Healthcare & Telemedicine

Remote Patient Vitals Alert & Monitoring App

Problem Context

A continuous monitoring pipeline for post-operative patients that ingests vital telemetry (SpO2, Heart Rate, BP), detects threshold breaches, and escalates emergency alerts through a caregiver matrix.

Target Stakeholders / Actors: Remote Patient, On-Call Caregiver

Requirements Table

Functional Requirements

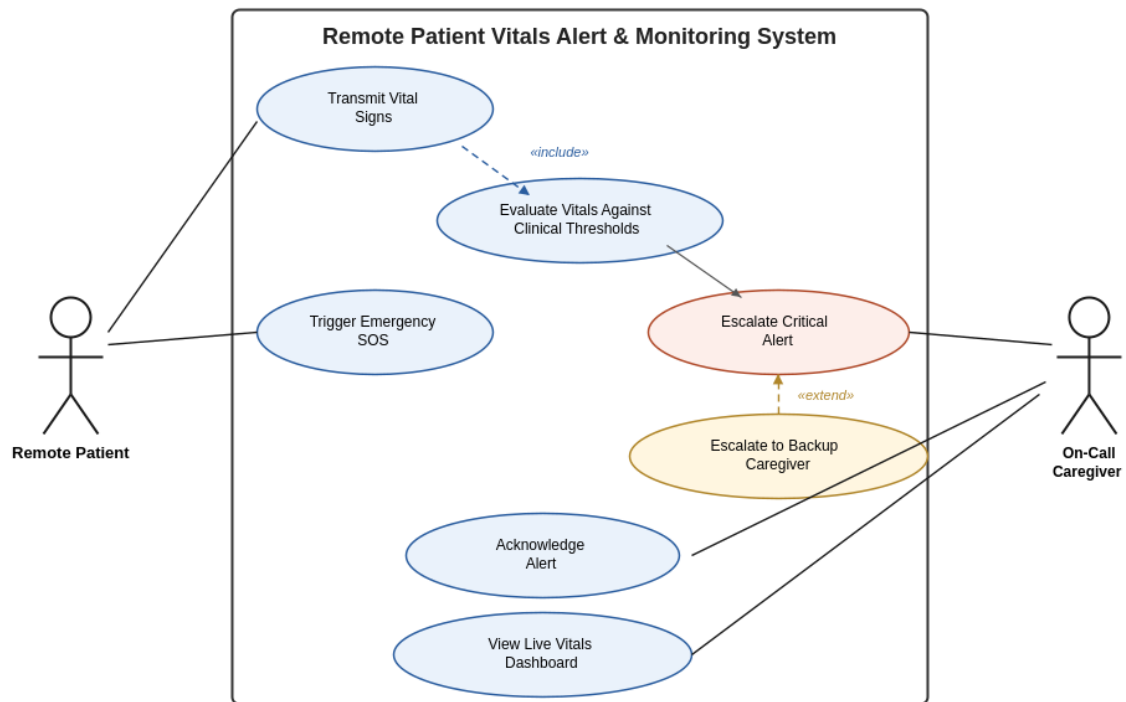
ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall continuously evaluate ingested patient vitals (SpO2, Heart Rate, BP) against configured clinical thresholds and flag critical anomalies within 2 seconds.	High	Pass: High heart rate (>140 BPM) triggers immediate escalation alert. Fail: Metric spike ignored or delayed > 5s.	Rapid anomaly detection is the core safety guarantee of a post-operative remote monitoring system; delayed detection directly endangers the patient.
FR-002	Functional	The system shall ingest continuous SpO2, Heart Rate, and BP telemetry streamed from the Remote Patient's wearable/monitoring device and timestamp each reading on receipt.	High	Pass: Every received reading is persisted with device ID and UTC timestamp within 1 second of arrival. Fail: A reading is dropped, unordered, or stored without a timestamp.	Accurate, ordered ingestion is the foundation all downstream threshold evaluation, alerting, and reporting depend on.
FR-003	Functional	The system shall escalate a critical alert to the On-Call Caregiver matrix (primary caregiver first, then backup tiers) whenever a threshold breach is detected, and shall automatically re-route to the next tier if the alert is not acknowledged within a configured time window.	High	Pass: Unacknowledged alert auto-escalates to backup caregiver within the configured window (e.g., 90 seconds). Fail: Alert remains only with an unresponsive primary caregiver with no re-routing.	A single unresponsive caregiver must never become a single point of failure for a life-critical alert.
FR-004	Functional	The system shall allow an authenticated On-Call Caregiver to acknowledge an active alert and to view the patient's real-time vitals and recent trend history from a monitoring dashboard.	Medium	Pass: Acknowledgement is timestamped, stops further escalation, and the dashboard reflects live vitals within 2 seconds. Fail: Acknowledged alert continues escalating, or dashboard shows stale (>10s old) data.	Caregivers need situational context and a reliable acknowledgement mechanism to respond appropriately and to stop redundant escalations.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-005	Functional	The system shall allow the Remote Patient to manually trigger an Emergency SOS from the patient app, which bypasses threshold evaluation and immediately raises a critical alert to the caregiver matrix.	Medium	Pass: SOS alert reaches the primary On-Call Caregiver within 2 seconds of activation. Fail: SOS is queued behind, or filtered by, routine threshold evaluation logic.	Patients may recognize a personal emergency before automated thresholds detect one, so a direct manual override is a required safety net.

Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	The telemetry ingestion gateway shall support at least 500 concurrent continuous telemetry streams with 99.99% uptime.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. Fail: Latency or availability degrades beyond target under peak concurrent load.	The platform must scale reliably across a growing post-operative patient population without dropping vitals or delaying anomaly detection.
NFR-002	Reliability & Data Privacy	All patient vitals and alert data shall be encrypted in transit and at rest (TLS 1.2+ / AES-256), and the alert-delivery path shall maintain redundant notification channels (push, SMS, voice call) so a critical alert is delivered within 30 seconds even if one channel fails.	High	Pass: Simulated failure of one notification channel still delivers the alert via a fallback channel within 30 seconds; all stored/transmitted data is confirmed encrypted. Fail: A single channel failure results in a missed or delayed critical alert, or any data is found unencrypted.	Healthcare telemetry is highly sensitive (HIPAA-class data), and alert delivery is life-critical, so both confidentiality and channel redundancy are non-negotiable.

UML Use-Case Diagram



Relationships modeled:

- «include» — Transmit Vital Signs includes Evaluate Vitals Against Clinical Thresholds (threshold evaluation is a mandatory step of every telemetry transmission).
- «extend» — Escalate to Backup Caregiver extends Escalate Critical Alert (backup escalation is an optional path that only fires if the primary caregiver does not acknowledge within the configured time window).

Use-Case Flow Specification

Use Case: Transmit Vital Signs

Primary Actor: Remote Patient (wearable / monitoring device)

Secondary Actor: On-Call Caregiver

Includes: Evaluate Vitals Against Clinical Thresholds

Related: Escalate Critical Alert is extended by Escalate to Backup Caregiver

Preconditions

- The Remote Patient is registered on the platform and their wearable/monitoring device is paired and actively transmitting.
- The On-Call Caregiver matrix (primary and backup tiers) is configured for the patient.
- The telemetry ingestion gateway is operational and reachable over the network.

Main Success Scenario

1. The Remote Patient's wearable device continuously transmits vitals (SpO2, Heart Rate, BP) to the platform.
2. The system validates and timestamps the incoming telemetry reading.
3. The system includes Evaluate Vitals Against Clinical Thresholds to check the ingested vitals against configured clinical thresholds.
4. The system determines no threshold breach has occurred; vitals are within the normal clinical range.
5. The system persists the reading to the patient's vitals history.
6. The system updates the On-Call Caregiver's live monitoring dashboard within 2 seconds (FR-001).
7. The use case ends successfully.

Alternate Flow

3a. Critical Anomaly Detected

1. During step 3, the system detects that a vital reading has breached a configured clinical threshold (e.g., Heart Rate > 140 BPM).
2. The system immediately triggers Escalate Critical Alert, notifying the primary On-Call Caregiver with the patient's vitals snapshot and location.
3. If the primary caregiver does not acknowledge the alert within the configured time window, the extend relationship fires Escalate to Backup Caregiver, notifying the next tier.
4. Once acknowledged (by primary or backup), escalation stops and processing resumes at step 5.

Postconditions

- The vitals reading has been validated, evaluated, and persisted to the patient's history.
- If a threshold breach occurred, a critical alert has been logged with its escalation path and final acknowledgement timestamp.
- The On-Call Caregiver dashboard reflects the patient's current status within the 2-second FR-001 target.